

29 Which statement best explains why the $[\text{TiCl}_6]^{2-}$ complex ion is expected to be colourless?

- A** The 3d subshell of the transition metal ion is empty.
- B** Electrons from the lower energy level absorb energy outside the visible spectrum.
- C** There is a large energy gap between the non-degenerate orbitals.
- D** There is no d-orbital splitting in the transition metal ion.